Statistical Analysis and Mortality Prediction of Heart Failure Clinical Records

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1 Motivation

Heart failure represents a major global health burden. Early detection and the management of high-risk patients are crucial to improving survival rates. Clinical datasets contain numerous physiological and biochemical indicators, such as patient age, ejection fraction, serum sodium, and creatinine phosphokinase (CPK) levels. However, not all indicators contribute equally to patient prognosis.

The primary motivation of this project is to leverage statistical learning techniques to filter out noise, reduce dimensionality, and identify the most significant predictive biomarkers for heart failure mortality. By understanding which features hold the most predictive power, medical professionals can prioritize specific diagnostic tests and tailor treatments to high-risk individuals. Furthermore, this project serves as a practical exercise in implementing core statistical algorithms (such as manual PCA and stepwise feature selection) from fundamental mathematical principles rather than solely relying on black-box machine learning libraries.

2 Implementation Details and Challenges

2.1 Implementation Details

The analysis pipeline was constructed using standard data science workflows:

- **Exploratory Data Analysis (EDA):** Initial univariate and bivariate analyses were conducted. Pairplots were generated to visualize the underlying distributions and pairwise correlations of continuous variables.
- **Principal Component Analysis (PCA):** We implemented PCA manually by constructing the centered data matrix, computing the covariance matrix, and solving for the eigenvalues and eigenvectors. This manual calculation was verified against built-in algorithms to ensure robust dimensionality reduction.
- **Linear Discriminant Analysis (LDA):** To classify survival outcomes, we trained LDA models under three conditions: using all standardized features (L1), using the top two principal components (L2), and using a manually selected subset of four features (L3). We evaluated the models using Leave-One-Out Cross-Validation (LOOCV).

- **Stepwise Regression:** We implemented a custom forward-backward stepwise regression algorithm utilizing F-tests to iteratively add or remove variables. This was compared against a full linear model.

2.2 Implementation Challenges

Several technical and statistical challenges were encountered during implementation:

- **Class Imbalance:** The dataset is highly imbalanced regarding the target variable (approx. 203 survivals vs. 96 deaths). This natural skew makes it difficult for algorithms like LDA to accurately classify the minority class without bias.

- **Low Variance Capture in PCA:** Calculating the eigenvalues revealed that the top principal components did not capture a dominant percentage of the variance (the first two captured only 33% combined). This challenge indicated that the dataset's features are fairly orthogonal, and aggressive dimensionality reduction would inherently lose predictive information.

3 Computational Results

3.1 Exploratory Data Analysis (EDA)

The bivariate analysis and correlation matrix revealed several insights. Age, serum creatinine, and time showed moderate correlations with the target variable. The generated scatter plot matrix (see Figure 1) demonstrated substantial overlap between feature distributions, hinting that simple linear separation in 2D space would be insufficient.

Figure 1: Scatter plot matrix for continuous variables.

3.2 Principal Component Analysis (PCA)

The manual eigenvalue decomposition produced a scree plot (Figure 2) that lacked a distinct "elbow," reinforcing the complexity of the clinical data. The explained variance ratio for the first two components was 17.26% and 16.11%, respectively. Because the cumulative variance of

the first two components was low, PCA was ultimately deemed ineffective for creating a highly accurate 2D predictive model.

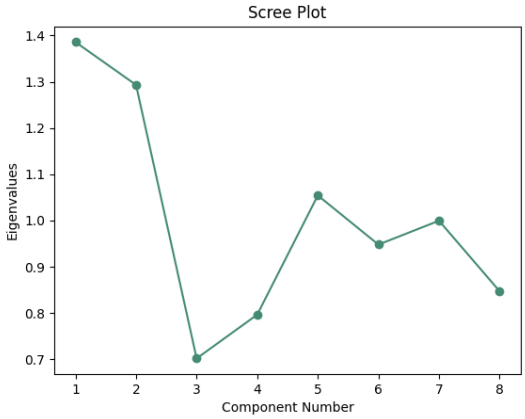


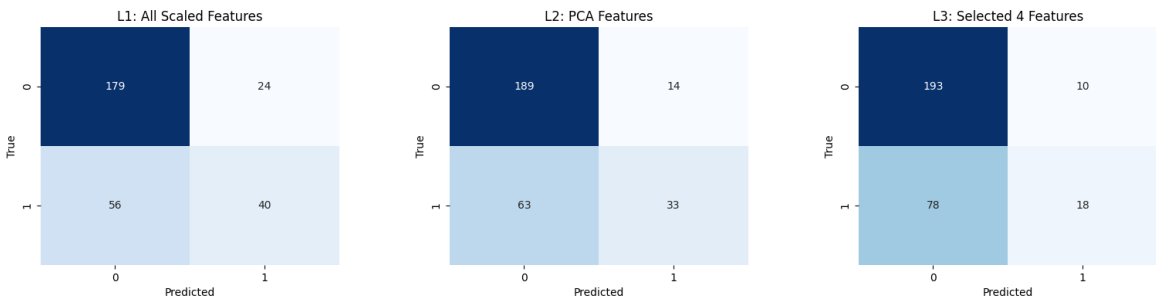
Figure 2: Scree plot displaying eigenvalues across principal components.

3.3 Linear Discriminant Analysis Classification

Using LOOCV, we compared the confusion matrices of three LDA models.

- **L1 (All Features):** Correctly identified 179 True Negatives and 40 True Positives (Figure 3a).
- **L2 (PCA Features):** Correctly identified 189 True Negatives but only 33 True Positives (Figure 3b). The reduction in true positives confirms that critical variance was lost during PCA.
- **L3 (Selected Features):** Yielded 193 True Negatives but dropped to 18 True Positives (Figure 3c), demonstrating high specificity but extremely poor sensitivity.

Overall, the full feature set (L1) maintained the best balance in detecting mortality events.



(a) L1: All Features (b) L2: PCA Features (c) L3: Selected Features

Figure 3: Confusion Matrices for LDA Models under LOOCV.

3.4 Regression and Feature Selection

The OLS regression model on the scaled dataset yielded an R^2 of 0.232 and an adjusted R^2 of 0.211. Despite the seemingly low R^2 (which is typical in complex biological and clinical data),

the F-statistic (10.97, $p < 0.001$) indicated that the model was highly significant overall.

According to the model parameters, the most statistically significant predictors ($p < 0.05$) were age, ejection rate, serum creatinine and serum sodium. Figure 4 visualizes the predicted scores of the full model (Y1) versus the stepwise model (Y2). While both models effectively assign higher risk scores to actual death events (colored dots), there remains an area of overlap, suggesting that clinical outcomes are influenced by unmeasured variables or nonlinear interactions.

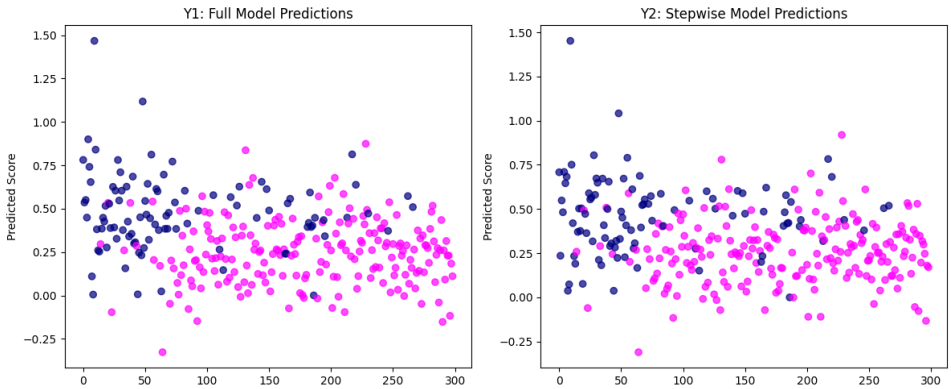


Figure 4: Comparison of predicted scores between Full Model (Y1) and Stepwise Model (Y2).

4 Future Directions

Based on the challenges and results documented above, future work on this dataset will focus on the following directions:

1. **Addressing Class Imbalance:** Implementing Synthetic Minority Over-sampling Technique (SMOTE) or adjusted class weighting could force the algorithms to penalize false negatives more heavily.

2. **Non-linear Modeling:** Because PCA demonstrated that linear combinations of features do not capture the data's complexity well, non-linear algorithms such as Random Forests, Support Vector Machines (with RBF kernels), or Gradient Boosting (XGBoost) should be explored.

3. **Survival Analysis:** Since the dataset includes a *time* variable (follow-up period), transitioning from binary classification to Cox Proportional-Hazards models (Survival Analysis) would provide a more nuanced understanding of patient risk over time.

5 Conclusion

This study successfully implemented statistical pipelines to analyze heart failure clinical records. Through manual calculations of PCA and stepwise regression, coupled with LDA cross-validation, we confirmed that age, ejection fraction, and serum creatinine are the most critical predictors of mortality. While linear models provide strong interpretability, future work leveraging non-linear techniques and survival analysis will be necessary to achieve medical-grade predictive accuracy.